AI / ML TRAINING PROGRAMME • DAY 2

# Multi-Class Classification

Machine Learning Training Report | Hemanth Gorijala | AI/ML Trainee

| 150 | ~97% | 3 | 5 |
|---|---|---|---|
| Dataset Observations | Model Accuracy | Target Classes | Pipeline Stages |

## Executive Summary

This report presents the implementation of a multi-class classification model using the Iris dataset as part of Day 2 training. The primary objective was to design and deploy a machine learning pipeline capable of handling multi-class predictions, evaluated against industry-standard performance metrics.

A Logistic Regression model was developed using a scikit-learn pipeline, ensuring a structured, reproducible, and reusable workflow. The model achieved high classification accuracy and was evaluated using a comprehensive suite of metrics including confusion matrix, precision, recall, F1-score, and ROC-AUC curves. The trained model was serialised as a .pkl file to support future deployment and reuse.

## Objectives

- Develop a multi-class classification model using a structured pipeline approach
- Understand the design and utility of machine learning pipelines in production-ready workflows
- Evaluate model performance using multiple industry-standard metrics
- Serialise the trained model for reuse and downstream deployment

## Dataset Overview

The Iris dataset is one of the most widely used benchmark datasets for multi-class classification. It consists of 150 labelled observations across three flower species, with four numerical features describing each sample. The dataset is clean, balanced, and contains no missing values — making it an ideal foundation for pipeline development and evaluation.

| Attribute | Details |
|---|---|
| **Total Records** | 150 observations |
| **Number of Features** | 4 numerical features (Sepal Length, Sepal Width, Petal Length, Petal Width) |
| **Target Variable** | Species (3 classes: Setosa, Versicolor, Virginica) |
| **Missing Values** | None — dataset is clean and complete |
| **Source** | scikit-learn built-in datasets (sklearn.datasets.load_iris) |

## Methodology

The project followed a structured five-stage pipeline, ensuring repeatability and clarity at each step of the machine learning workflow.

| # | Stage | Actions Performed |
|---|---|---|
| 1 | **Data Loading** | Dataset loaded using sklearn.datasets.load_iris() and converted into a structured pandas DataFrame for processing. |
| 2 | **Data Splitting** | Dataset split 80/20 into training (120 samples) and testing (30 samples) sets using train_test_split() for unbiased evaluation. |
| 3 | **Pipeline Development** | A scikit-learn Pipeline was constructed combining StandardScaler (feature normalisation) and LogisticRegression (classification). |
| 4 | **Model Training** | Pipeline fitted on the training dataset. StandardScaler computed mean/std; Logistic Regression learned decision boundaries. |
| 5 | **Evaluation & Persistence** | Model evaluated with accuracy, confusion matrix, precision, recall, F1-score, and ROC-AUC. Saved as iris_model.pkl for deployment. |

## Model Evaluation

The model was evaluated using a comprehensive set of performance metrics to assess its classification capability across all three target classes. Each metric provides a different lens on model quality, collectively offering a complete picture of predictive performance.

| Metric | Description | Observed Result |
|---|---|---|
| **Accuracy** | Proportion of correctly classified instances | **~95% – 100%** |
| **Confusion Matrix** | Grid of actual vs. predicted values across all 3 classes | Minimal misclassifications |
| **Precision** | Correctness of positive predictions per class | High across all classes |
| **Recall** | Completeness of predictions — true positive rate per class | High across all classes |
| **F1 Score** | Harmonic mean of Precision and Recall | Balanced and strong |
| **ROC-AUC** | One-vs-Rest multi-class curve measuring class separability | Strong separation per class |

### 5.1 Accuracy

The model achieved an accuracy between 95% and 100% on the test set, demonstrating strong generalisation to unseen data. The high accuracy reflects the well-defined feature boundaries present in the Iris dataset.

### 5.2 Confusion Matrix

The confusion matrix provided a granular breakdown of correct and incorrect predictions across Setosa, Versicolor, and Virginica. Very few misclassifications were observed, primarily between the closely related Versicolor and Virginica classes — which share overlapping feature distributions.

### 5.3 Precision, Recall & F1-Score

Precision, Recall, and F1-Score were computed per class. All three metrics returned high values, confirming that the model predicts each class both correctly and completely. The F1-Score, as the harmonic mean of Precision and Recall, confirms a well-balanced classifier with no significant trade-offs between false positives and false negatives.

### 5.4 ROC-AUC Curve

ROC-AUC analysis was implemented using the One-vs-Rest (OvR) strategy, generating a separate curve for each class. The AUC scores were consistently strong, confirming excellent class separability and robust discriminative power across all three species.

## Model Persistence

Upon successful training and evaluation, the complete pipeline (including the fitted scaler and classifier) was serialised and saved to disk for future use.

| Property | Value |
|---|---|
| **File Name** | iris_model.pkl |
| **Format** | Python Pickle (.pkl) via joblib serialisation |
| **Contents** | Full pipeline: StandardScaler + fitted LogisticRegression model |
| **Use Cases** | Reuse without retraining; deployment into real-world applications |

## Key Findings

| Finding | Implication for Next Steps |
|---|---|
| **High model accuracy achieved (~97%)** | Logistic Regression is highly effective on linearly separable data like Iris. Suitable as a strong baseline. |
| **Minimal misclassifications observed** | Most errors likely between Versicolor and Virginica — expected due to overlapping feature distributions. |
| **Pipeline simplifies workflow** | Reusable pipeline architecture can be extended with new preprocessing steps or classifiers without restructuring. |
| **Model saved as .pkl file** | Enables deployment and reuse without retraining. Supports integration into downstream applications. |

## Conclusion

This project successfully demonstrated the end-to-end implementation of a multi-class classification pipeline using the Iris dataset. The structured pipeline approach — combining feature scaling with logistic regression — yielded high accuracy and strong performance across all evaluation metrics.

The ability to serialise the trained pipeline as a .pkl file significantly enhances its practical usability, enabling seamless reuse and deployment without retraining. The findings confirm that Logistic Regression, when paired with appropriate preprocessing, serves as a highly effective baseline classifier for linearly separable datasets.